Precision Envelope Targeting of Neural Oscillation Bands: Exact Rhythmic Formulas, a Conflict-Coherence Framework, and a Unique Carrier Theorem for the Theta Band

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Abstract

We present a conditional theoretical framework for how music could interact with neural oscillation bands through low-frequency amplitude envelopes. The framework is conditioned on a falsifiable seven-mode hypothesis: neural oscillation timing has a period-8 structure (motivated by the minimal neutral cycle on a 3-dimensional binary hypercube) whose DFT produces seven non-DC modes with a rigid 3+1+3 conjugate-pair architecture. With one calibration ($f_0 = 40$ Hz), the mode frequencies become $5, 10, \ldots, 35$ Hz. From this conditional model we derive four results. *First*, ordinary musical structures—rhythm, beating, and tremolo—generate amplitude envelopes at calculable frequencies; for every mode there exists a specific tempo-subdivision pair that targets it exactly. *Second*, a cost function $J(r) = \frac{1}{2}(r + r^{-1}) - 1$ on frequency ratios feeds into a phenomenological conflict-coherence model: lower aggregate cost yields higher coherence and, within a proxy strain model, larger predicted strain reduction. *Third*, mode 1 (theta, 5 Hz) is the unique full-ladder carrier—the only mode whose integer harmonics span all seven modes—predicting that theta stability is a structural prerequisite for higher-band coherence. *Fourth*, the gamma/theta nesting ratio $f_7/f_1 = 7$ is a parameter-free consequence of the period alone within the model, consistent with the empirically reported $\approx$7:1 gamma-to-theta nesting. All structural results are parameter-free within the model; Hz values require only the single calibration. The core derivations have machine-checked proofs in Lean 4, while the empirical band identifications and therapeutic claims remain to be tested. We present eight falsifiable predictions together with proposed preregistration anchors for refutation.

Keywords. Neural oscillations, theta-gamma coupling, music therapy, amplitude envelope, auditory steady-state response, formal verification, cross-frequency coupling, EEG

1 Introduction

Music therapy has well-documented effects on mood, pain perception, cognitive function, and neurological rehabilitation [Thaut & Hoemberg, 2014, Bradt et al., 2021]. Electroencephalographic (EEG) studies consistently show that musical stimuli modulate neural oscillation power and cross-frequency coupling, with theta-gamma coupling receiving particular attention as a substrate for memory encoding and attentional control [Lisman & Jensen, 2013, Canolty & Knight, 2010]. Auditory steady-state responses (ASSRs) demonstrate that amplitude-modulated sound can entrain neural oscillations at the modulation frequency [Galambos et al., 1981, Picton et al., 2003].

Despite these empirical foundations, a unifying theoretical framework is lacking. To our knowledge, existing models describe *which* frequencies entrain and *that* music affects neural state, but do not provide:

(a) A structural reason why specific frequency bands (theta, alpha, beta, gamma) are privileged rather than arbitrary empirical categories;

(b) Exact formulas connecting musical tempo, subdivision, and modulation to specific target frequencies;

(c) A quantitative measure of musical "conflict" or "coherence" that predicts therapeutic efficacy;

(d) A structural explanation for the empirically observed privileged role of the theta band in cross-frequency coupling.

This paper proposes a candidate framework aimed at all four gaps. Its claims are explicitly conditional: if neural timing is well-modeled by the eight-slot structure of Hypothesis 1, then the envelope-driving formulas, carrier hierarchy, and nesting ratio follow. We begin with that single falsifiable hypothesis about the periodic structure of neural oscillation timing (Section 2). From it we derive an envelope-driving mechanism (Section 3), a cost-based conflict-coherence model (Section 4), and a unique carrier theorem for the theta band (Section 5). Section 6 presents eight testable predictions and suggested study-design anchors for falsification. Section 7 summarizes the machine-checked support for the core derivations. Section 8 discusses limitations, comparisons with existing theories, and future directions.

2 The Seven-Mode Hypothesis

Hypothesis 1 (Eight-Slot Periodic Structure)**.** *Neural oscillation timing possesses a minimal periodic structure with period $N = 8$. The seven non-trivial (non-DC) modes of this structure define a discrete frequency ladder at integer multiples of a base frequency f_0.*

Motivation. The period $N = 8 = 2^3$ can be motivated from several directions. *Binary-walk argument:* a system whose state is described by $D = 3$ independent binary variables has $2^3 = 8$ possible configurations; the minimal neutral cycle (a closed walk that visits every vertex of the 3-cube with zero net displacement on each axis) requires exactly 8 steps. Three is the lowest dimension admitting non-trivial topological linking of closed curves [Rolfsen, 1976], which is relevant whenever a system must support topologically protected conserved quantities. *Empirical convergence:* (i) the ubiquity of octave (factor-of-2) structure in auditory processing, (ii) the existence of approximately seven canonical EEG frequency bands from theta through high gamma, and (iii) the empirical success of 40 Hz stimulation protocols [Iaccarino et al., 2016], suggesting 40 Hz as a natural ceiling frequency, collectively motivate testing $N = 8$. The present paper does not require commitment to any particular derivation of $N = 8$; Hypothesis 1 is offered as a standalone, falsifiable model. The contribution of the paper is to make its downstream consequences explicit and experimentally vulnerable.

Provenance convention. Each claim in this paper carries one of four labels. A **core theorem** depends only on formal definitions internal to the model (including, where relevant, $N = 8$ and the DFT structure), with no empirical calibration or data. A **calibrated corollary** additionally requires $f_{\text{cycle}} = 40$ Hz. An **empirical identification** maps a derived quantity onto an independently measured phenomenon. A **hypothesis** is a testable claim whose failure would not invalidate the core derivation. Table 7 in Section 8 records the status of every claim.

DFT-8 structure. The Discrete Fourier Transform of an N-point periodic signal has N frequency bins indexed $k = 0, \ldots, N - 1$. For $N = 8$:

- Bin $k = 0$ is the DC (mean) component.

- Bins $k = 1$ through $k = 7$ are the seven non-DC modes.
- Conjugate symmetry of real signals pairs bins k and $8-k$: modes $(1,7)$, $(2,6)$, $(3,5)$ carry conjugate information.
- Bin $k = 4$ (Nyquist) is the unique self-conjugate non-DC mode.

This gives a $3+1+3$ mirror structure: three sub-Nyquist modes, one Nyquist mode, and three super-Nyquist modes.

Frequency assignment. If the full 8-slot cycle completes at rate $f_{\text{cycle}} = 40$ Hz, mode k has frequency

$$f_k = \frac{k \cdot f_{\text{cycle}}}{N} = 5k \quad \text{Hz}, \qquad k = 1, \ldots, 7. \tag{1}$$

The resulting ladder is:

Table 1: Seven-mode frequency ladder and EEG band correspondence.

Mode k	Frequency (Hz)	EEG Band	Conjugate
1	5	Theta	7
2	10	Alpha	6
3	15	Low Beta	5
4	20	Beta (Nyquist)	self
5	25	High Beta	3
6	30	Low Gamma	2
7	35	Gamma	1

Proposition 1 (Strict mode ordering)**.** *If $1 \le i < j \le 7$, then $f_i < f_j$.*

Proof. Equation (1) gives $f_i = 5i$ and $f_j = 5j$; since $5 > 0$, the inequality $i < j$ implies $f_i < f_j$. □

Falsification criterion. Hypothesis 1 predicts that source-resolved or task-resolved spectral decompositions, and especially entrainment experiments, should reveal preferred response maxima at or near $5, 10, 15, 20, 25, 30, 35$ Hz rather than at arbitrary offsets. The conjugate structure further predicts enhanced cross-frequency phase coupling between paired modes (1↔7, 2↔6, 3↔5).

3 Envelope Driving: How Music Reaches the Mode Ladder

Audible musical pitch (roughly 100–4000 Hz) is far above the 5–35 Hz mode ladder and cannot directly entrain these oscillations. However, music naturally generates low-frequency *amplitude envelopes* through three mechanisms.

3.1 Rhythmic envelopes

A musical performance at tempo T beats per minute (BPM) with rhythmic subdivision s (e.g., $s = 2$ for eighth notes, $s = 4$ for sixteenths) produces amplitude fluctuations at frequency

$$f_{\text{env}} = \frac{T \cdot s}{60} \quad \text{Hz}. \tag{2}$$

Definition 2 (Target tempo)**.** *For mode $k \in \{1, \ldots, 7\}$ and subdivision $s > 0$, the* target tempo *is the BPM at which the rhythmic envelope exactly matches mode k:*

$$T_k(s) = \frac{60 \cdot f_k}{s} = \frac{300k}{s} \quad \textit{BPM}. \tag{3}$$

Theorem 3 (Exact rhythmic mode targeting)**.** *For every mode $k \in \{1, \ldots, 7\}$ and every positive subdivision s, playing music at tempo $T_k(s)$ with subdivision s produces an envelope frequency equal to exactly $f_k = 5k$ Hz.*

Proof. Direct substitution: $f_{\text{env}} = T_k(s) \cdot s/60 = (300k/s) \cdot s/60 = 5k = f_k$. □

In the companion Lean formalization this calculation is formalized in the envelope-driving module. Table 2 gives concrete examples.

Table 2: Target tempos for mode-specific entrainment via rhythmic subdivision. Each cell gives the BPM at which subdivision s produces an envelope at exactly f_k Hz.

	$s = 2$	$s = 4$	$s = 6$	$s = 8$	$s = 16$
Mode 1 (5 Hz)	150	75	50	37.5	18.75
Mode 2 (10 Hz)	300	150	100	75	37.5
Mode 3 (15 Hz)	450	225	150	112.5	56.25
Mode 4 (20 Hz)	600	300	200	150	75
Mode 5 (25 Hz)	750	375	250	187.5	93.75
Mode 6 (30 Hz)	900	450	300	225	112.5
Mode 7 (35 Hz)	1050	525	350	262.5	131.25

Musically realistic tempos (60–200 BPM) with standard subdivisions ($s = 4$ to $s = 16$) cover modes 1 through 4 directly. Higher modes require either fast tempos with large subdivisions or one of the additional mechanisms below.

3.2 Beat envelopes

Two simultaneous tones at frequencies f_1 and f_2 produce an amplitude modulation ("beating") at frequency

$$f_{\text{beat}} = |f_1 - f_2|. \tag{4}$$

Theorem 4 (Beat-envelope mode targeting)**.** *For any carrier frequency $f_c > 0$ and mode k, the tone pair $(f_c,\ f_c + f_k)$ produces a beat envelope at exactly f_k.*

This is the mechanism exploited by binaural-beat protocols [Oster, 1973]. The present framework specifies *which* beat frequencies should be most effective: those at exactly $5k$ Hz for $k = 1, \ldots, 7$.

3.3 Tremolo and amplitude modulation

Tremolo (periodic amplitude variation) at rate f_{mod} directly generates an envelope at that frequency. Vibrato (periodic pitch variation) generates sidebands whose beating produces additional low-frequency envelopes. In both cases the envelope frequency can be chosen to match a target mode.

3.4 Summary of envelope channels

Table 3: Three channels by which music generates low-frequency envelopes in the 5–35 Hz range.

Channel	Formula	Control parameter
Rhythm	$f_{\text{env}} = Ts/60$	Tempo T, subdivision s
Beating	$f_{\text{env}} = \lvert f_1 - f_2 \rvert$	Frequency separation
Tremolo	$f_{\text{env}} = f_{\text{mod}}$	Modulation rate

4 The Cost-Conflict-Coherence Framework

The previous section shows *how* envelopes reach the mode ladder. This section addresses *which combinations of tones* are most promising within the model, by introducing a quantitative measure of interval and chord conflict.

4.1 Interval cost

Definition 5 (Interval cost function)**.** *For a frequency ratio $r > 0$, define*

$$J(r) = \tfrac{1}{2}\big(r + r^{-1}\big) - 1. \tag{5}$$

The function J has the following properties:

(i) $J(1) = 0$ (unison has zero cost).

(ii) $J(r) = J(r^{-1})$ (symmetric under inversion).

(iii) $J(r) > 0$ for all $r \neq 1$.

(iv) J is strictly convex on $(0, \infty)$.

The function J can be derived axiomatically as the unique binary-symmetric, identity-anchored cost on positive reals satisfying a multiplicative d'Alembert functional equation. For the purposes of this paper, it may simply be taken as a candidate consonance measure and evaluated against psychoacoustic data.

4.2 The interval-cost hierarchy

Applying J to standard just-intonation ratios yields:

Table 4: J-cost of standard musical intervals. Lower cost corresponds to lower conflict.

Interval	Ratio r	$J(r)$	Decimal
Unison	1/1	0	0
Minor third	6/5	1/60	0.0167
Major third	5/4	1/40	0.0250
Perfect fourth	4/3	1/24	0.0417
Tritone	45/32	169/2880	0.0587
Perfect fifth	3/2	1/12	0.0833
Octave	2/1	1/4	0.2500

Theorem 6 (Complete interval-cost ordering)**.**

$$J(1) < J\big(\tfrac{6}{5}\big) < J\big(\tfrac{5}{4}\big) < J\big(\tfrac{4}{3}\big) < J\big(\tfrac{45}{32}\big) < J\big(\tfrac{3}{2}\big) < J(2).$$

Notable feature. This ordering differs from traditional consonance rankings in a specific, testable way: the perfect fifth ($J = 1/12$) has *higher* cost than the minor third ($J = 1/60$), major third ($J = 1/40$), and perfect fourth ($J = 1/24$). In the present framework, the fifth's structural importance arises not from having the lowest cost, but from a different property: it is the ratio $12/8 = 3/2$ linking the 12-semitone chromatic octave to the 8-slot DFT basis. The twelve-fifths walk around the circle of fifths closes the octave with error less than 2% (the Pythagorean comma), making the fifth a structural bridge rather than a cost optimum.

4.3 Chord conflict and coherence

Definition 7 (Chord conflict)**.** *For a chord $C = \{f_1, \ldots, f_n\}$ with $f_i > 0$, the* chord conflict *is the directed aggregate of pairwise interval costs:*

$$\mathrm{Conflict}(C) = \sum_{i \neq j} J\left(\frac{f_i}{f_j}\right). \tag{6}$$

Because $J(r) = J(r^{-1})$, this directed aggregate is exactly twice the usual unordered pair sum when all notes are distinct. We use the directed form because it matches the companion Lean implementation exactly.

Definition 8 (Chord coherence)**.** *The* chord coherence *normalizes conflict to a score in* $(0, 1]$*:*

$$\mathrm{Coh}(C) = \frac{1}{1 + \mathrm{Conflict}(C)}. \tag{7}$$

Proposition 9 (Lower conflict implies higher coherence)**.** *If* $\mathrm{Conflict}(C_1) \leq \mathrm{Conflict}(C_2)$*, then* $\mathrm{Coh}(C_2) \leq \mathrm{Coh}(C_1)$*.*

4.4 Effective alignment and a phenomenological strain proxy

We introduce a minimal phenomenological model that converts the coherence ordering above into a rank-order prediction about downstream response. Let σ_0 denote a latent pre-stimulus strain proxy (deviation from mode-optimal oscillation), $d > 0$ the driving strength of the envelope, and $a > 0$ the base alignment (a measure of stimulus-mode phase match). The proxy σ is not claimed to be a directly observed biophysical state variable; rather, it stands in for a preregistered physiological or clinical endpoint expected to decrease as alignment improves. The *effective alignment* of a chord is

$$a_{\mathrm{eff}}(C) = a \cdot \mathrm{Coh}(C), \tag{8}$$

and the post-stimulus strain is

$$\sigma_1 = \sigma_0 \cdot \max\bigl(0,\ 1 - d \cdot a_{\mathrm{eff}}(C)\bigr). \tag{9}$$

Theorem 10 (Lower conflict lowers the strain proxy)**.** *For fixed $\sigma_0 \geq 0$, $d > 0$, $a \geq 0$, if* $\mathrm{Conflict}(C_1) \leq \mathrm{Conflict}(C_2)$*, then $\sigma_1(C_1) \leq \sigma_1(C_2)$.*

Proof. By Proposition 9, $\mathrm{Coh}(C_2) \leq \mathrm{Coh}(C_1)$, so $a_{\mathrm{eff}}(C_2) \leq a_{\mathrm{eff}}(C_1)$. Then $1 - d \cdot a_{\mathrm{eff}}(C_1) \leq 1 - d \cdot a_{\mathrm{eff}}(C_2)$, and applying $\max(0, \cdot)$ and multiplication by $\sigma_0 \geq 0$ preserves the inequality. □

Theorem 10 gives a rank-order prediction within this proxy model: *among stimuli that successfully drive the same target mode and are evaluated with the same preregistered endpoint, those with lower aggregate interval cost produce a lower post-stimulus strain proxy.* Whether that proxy maps to clinically meaningful benefit remains an empirical question outside the present derivation.

5 The Unique Carrier-Wave Property of the Theta Band

Definition 11 (Harmonic support)**.** *Mode b* harmonically supports *mode t if $f_t = n \cdot f_b$ for some positive integer n.*

Definition 12 (Full-ladder carrier)**.** *Mode b is a* full-ladder carrier *if it harmonically supports every mode $t \in \{1, \ldots, 7\}$.*

Theorem 13 (Mode 1 is a full-ladder carrier)**.** *Mode 1 ($f_1 = 5$ Hz) harmonically supports every mode: $f_k = k \cdot f_1$ for $k = 1, \ldots, 7$.*

Proof. $f_k = 5k = k \cdot 5 = k \cdot f_1$, with multiplier $n = k > 0$. □

Theorem 14 (Mode 1 is the *unique* full-ladder carrier)**.** *For $b \in \{1, \ldots, 7\}$, mode b is a full-ladder carrier if and only if $b = 1$.*

Proof. The forward direction is Theorem 13. For the converse, suppose $b \geq 2$. Then $f_b = 5b \geq 10$, but $f_1 = 5 < 10 \leq 5b$. For mode b to support mode 1, we need $5 = n \cdot 5b$, i.e., $n = 1/b < 1$, which is not a positive integer. Hence mode b does not support mode 1 and is not a full-ladder carrier. □

Interpretation. Theorem 14 says that theta (5 Hz) is the *only* mode that can serve as a harmonic foundation for the entire oscillation ladder. Every higher mode's frequency is a whole-number multiple of theta, so a stable theta oscillation provides a phase reference against which every higher mode can lock. No higher mode reciprocates: mode 2 (10 Hz) cannot generate 5 Hz as an integer multiple; mode 3 (15 Hz) cannot generate 10 Hz; and so on.

This gives a structural prediction about directionality in cross-frequency interactions:

1. Disrupting theta should degrade coherence across all higher bands.
2. Disrupting gamma should leave lower bands substantially intact.
3. Entrainment protocols that establish theta first and then progressively engage higher modes should outperform protocols that target higher modes directly.

Existing literature on theta-gamma coupling [Lisman & Jensen, 2013, Canolty & Knight, 2010, Jensen & Colgin, 2007] and the role of theta in memory consolidation, navigation, and attentional gating is consistent with this structural prediction, but the *uniqueness* of theta's carrier role—that no other band can serve this function—has not previously been stated as a mathematical theorem.

5.1 The gamma/theta nesting ratio

The ratio of the highest to the lowest non-DC mode frequency is a parameter-free structural invariant:

Theorem 15 (Nesting ratio)**.** $f_7/f_1 = (N-1) \cdot f_0/(1 \cdot f_0) = N - 1 = 7$. *This ratio is independent of f_0.*

The nesting ratio of ≈ 7 gamma cycles per theta cycle has been observed in cross-frequency coupling studies [Lisman & Jensen, 2013, Canolty & Knight, 2010]. Within the DFT-8 model, this ratio is not tuned by free parameters: for *any* DFT-N system with $N = 8$, the highest-to-lowest non-DC frequency ratio is $N - 1 = 7$.

5.2 Conjugate coherence structure

Two modes share an oscillation family if and only if they are equal or conjugate. This is proved by exhaustive case analysis on all 7×7 mode pairs (Lean: `family_eq_iff_conjugate`). The structural consequence: within-family (conjugate-pair) coherence should exceed between-family coherence in EEG data, because conjugate DFT modes of a real-valued signal carry identical spectral magnitude.

6 Testable Predictions

The framework generates eight specific predictions. Table 5 records the measurement target and qualitative failure mode for each prediction; the numerical design anchors discussed below are proposed preregistration templates rather than universal evidentiary standards.

Table 5: Eight testable predictions derived from the seven-mode hypothesis, envelope-driving formulas, conflict-coherence model, and carrier theorem.

#	Prediction	Measure	Qualitative falsifier
1	Source-resolved or task-resolved EEG, and especially entrainment experiments, reveal response maxima near 5, 10, 15, 20, 25, 30, 35 Hz (± 0.5 Hz).	Power spectral density / evoked response maxima	Maxima absent or at substantially different frequencies
2	Binaural beats at on-ladder frequencies ($5k$ Hz) produce stronger ASSRs than beats at off-ladder frequencies ($5k \pm 1.5$ Hz).	ASSR amplitude at target frequency	No on-ladder vs. off-ladder difference
3	Music at the calculated target tempo $T_k(s)$ entrains mode k more strongly than music at a randomly chosen tempo.	EEG power at f_k during stimulation	No tempo-specific entrainment effect
4	The J-cost ordering predicts inter-band EEG coherence better than traditional consonance rankings (Helmholtz roughness, harmonic template).	Correlation between $J(r)$ rank and measured coherence, compared to alternative models	Traditional rankings correlate equally well or better
5	Theta-first sequential entrainment ($5 \to 10 \to 15 \to \cdots$ Hz) produces higher whole-spectrum coherence than gamma-first ($35 \to 30 \to \cdots$ Hz) or simultaneous protocols.	Average cross-frequency coherence; cognitive battery scores	No order effect, or reverse order is superior
6	Disrupting theta via TMS degrades cross-frequency coupling across all higher bands; disrupting gamma leaves lower bands intact.	Phase-amplitude coupling before and after disruption	Symmetric degradation in both directions
7	Heart rate variability (HRV) improvement correlates with J-cost rank of sustained intervals, not with traditional consonance rank.	HRV spectral power vs. interval J-cost	HRV tracks traditional consonance
8	Among chords matched for loudness and duration, those with lower aggregate conflict $\sum_{i \neq j} J(f_i/f_j)$ produce greater self-reported relaxation and lower EEG strain markers.	Conflict score vs. preregistered endpoint (e.g., VAS, EEG alpha power)	No correlation, or roughness-based models predict better

Experimental design notes. The numerical thresholds below should be read as proposed preregistration anchors. Their role is to make the model concretely refutable in prospective studies, not to claim that one sample size or significance threshold is optimal for every cohort, modality, or endpoint. Any actual study should recompute power from endpoint reliability and preregister one primary analysis. Predictions 1–3 require only standard EEG equipment (≥ 64 channels, resting-state or task-resolved) and audio generation software; the target-tempo formula (Eq. 3) provides exact stimulus parameters. For Prediction 1, a natural preregistration anchor is a peak-detection rule testing whether dominant response maxima lie within ± 0.5 Hz

of $5, 10, 15, 20, 25, 30, 35$ Hz and whether adjacent peaks are separated by approximately 5 Hz; systematic displacement from this ladder counts against the model. For Prediction 2, a natural anchor is a preregistered on-ladder versus off-ladder contrast using matched carriers, matched sound levels, and permutation-based inference; repeated null results or confidence intervals centered near zero in adequately powered cohorts would count against the model. Predictions 4, 7, and 8 compare J-cost against established consonance models (Helmholtz roughness, Plomp–Levelt critical bandwidth, harmonic template), permitting direct model comparison; falsified if traditional rankings correlate equally well or better with the dependent variable. Prediction 8 additionally requires a preregistered operationalization of the latent strain proxy in Eq. 9, for example a composite endpoint combining relaxation self-report with specified EEG markers. Predictions 5 and 6 test the carrier theorem's directional asymmetry and require either sequential stimulation protocols or focused TMS; falsified if disruption effects are symmetric across bands or if reverse-order protocols perform equally well.

7 Formal Verification

The core mathematical consequences used in Sections 2 through 5 have corresponding machine-checked proofs in the Lean 4 proof assistant [de Moura et al., 2021]. The cited modules build with zero `sorry` and zero `admit`. The verified statements cover internal consequences of the DFT-8 hypothesis and the definitions introduced in the paper; they do not, by themselves, validate the empirical band identifications, the 40 Hz calibration, or any therapeutic efficacy claim.

The key verified modules and their principal theorems are:

Table 6: Lean 4 modules and verified theorems corresponding to this paper's results.

Module	Key theorem(s)
`Consciousness.NeutralModeStructure`	`neutral_dim`, `dim_from_spatial`
	`mode_frequencies`, `freq_monotone`
	`nyquist_unique_self_conjugate`
	`family_eq_iff_conjugate`
	`conjugate_frequency_complement`
`Consciousness.ModeCarrier`	`fullLadderCarrier_iff_mode1`
	`sequential_activation_from_mode1`
`Consciousness.SevenModesCert`	`Cert.verified_instance` (bridge certificate)
`MusicTheory.EnvelopeDriving`	`targetTempoForMode_bpm_formula`
	`subdivision_envelope_drives_mode`
	`beatEnvelope_drives_mode`
	`tremoloEnvelope_drives_mode`
`MusicTheory.ModeConflict`	`lowerConflict_higherCoherence`
	`higherConflict_lower_alignment`
`MusicTheory.Consonance`	`extended_ratio_cost_hierarchy`
`Consciousness.SoundHealing`	`lowerConflict_reduces_strain_more`
	`chord_sound_reduces_strain`

The full source code is available at `github.com/jonwashburn/reality`. A machine-checkable bridge certificate (`SevenModesCert.lean`) chains every structural result from the period assumption through to the final predictions. All modules compile with Lean 4 / Mathlib with zero unproved obligations, ensuring that the model's key equations and monotonicity claims are deductively established from the stated definitions once the formal assumptions are granted. The formal verification should therefore be read as an internal consistency certificate, not as empirical validation of Hypothesis 1 or of the therapeutic application layer.

8 Discussion

8.1 Provenance matrix

Table 7 classifies every theorem-like statement.

Table 7: Provenance classification. **C** = core theorem ($N = 8$ only), **Cal** = calibrated corollary ($f_0 = 40$ Hz required), **E** = empirical identification, **H** = hypothesis.

Statement	**Location**	**Prov.**
7 non-DC modes	Hyp. 1	C
Conjugate pairing, 3+1+3	Sec. 2	C
Family ⇔ conjugacy	Sec. 5	C
Nesting ratio $f_7/f_1 = 7$	Thm 15	C
Unique carrier (mode 1)	Thm 14	C
Freq. monotonicity	Prop. 1	Cal
Lower conflict ⇒ lower strain proxy	Thm 10	C
Frequency ladder Hz	Eq. 1	Cal
Target tempos	Def. 2	Cal
EEG band identification	Table 1	E
Eight-slot hypothesis	Hyp. 1	H
Predictions 1–8	Table 5	H

8.2 Relationship to existing theories

Auditory steady-state responses. The ASSR literature [Galambos et al., 1981, Picton et al., 2003, Gao et al., 2014] has established that amplitude-modulated sound entrains neural oscillations. The present framework is consistent with these findings and adds specificity: within the model it predicts that entrainment should be strongest at the seven on-ladder frequencies, provides exact formulas for generating those frequencies from standard musical structures, and offers a cost function for ranking candidate stimuli.

Theta-gamma coupling. The theta-gamma coupling literature [Lisman & Jensen, 2013, Canolty & Knight, 2010] documents a particularly strong and functionally important form of cross-frequency interaction. The carrier theorem (Theorem 14) provides a candidate structural explanation within the model: theta is the unique mode whose harmonics span the full ladder, making it the natural phase reference for all higher oscillations. The nesting ratio $f_7/f_1 = 7$ (Theorem 15) matches the empirically observed $\approx$ 7:1 gamma-to-theta nesting and is, within the model, a parameter-free consequence of the period. If the model is correct, this predicts not just that theta-gamma coupling exists, but that theta's role is *uniquely privileged*—a claim testable by comparing disruption effects across bands (Prediction 6).

Communication through coherence. The CTC framework [Fries, 2015] explains *why* oscillatory coherence matters for neural communication (coherent oscillations create windows of effective connectivity) but does not predict the specific band count or frequency ratios. The DFT-8 model is complementary: it derives the mode *structure* (count, pairing, carrier hierarchy) from a symmetry assumption, while CTC explains the functional significance of coherence between those modes.

Mechanistic models. Thalamocortical loop models [Lopes da Silva, 2013] derive oscillation frequencies from biophysical parameters (membrane time constants, axonal delays), and interneuron-network models [Kopell et al., 2011] provide detailed accounts of beta and gamma

generation. These models explain band *generation* but not band *count*. The present approach is again complementary: the DFT-8 structure constrains which modes are possible; biophysical models explain how specific circuits instantiate them.

Consonance models. The J-cost function differs from Helmholtz roughness [Helmholtz, 1863], Plomp-Levelt critical bandwidth models [Plomp & Levelt, 1965], and harmonic template models [Parncutt & Hair, 2011] in its ordering of intervals. Specifically, J-cost ranks the minor third as less conflicting than the perfect fifth, whereas traditional models typically rank the fifth as more consonant. Predictions 4, 7, and 8 are designed to adjudicate between these models empirically.

8.3 Limitations

1. **The auditory demodulation pathway is not specified.** The framework shows that musical structures *contain* envelopes at the right frequencies but does not model the neural mechanism by which the auditory system extracts and transmits those envelopes to oscillatory circuits. This is a gap that existing ASSR physiology partially fills [Picton et al., 2003].

2. **The seven-mode hypothesis is a hypothesis.** While motivated by the DFT-8 structure and consistent with known EEG bands, the claim that neural timing has an exact 8-slot period at 40 Hz has not been independently established. Prediction 1 is therefore best interpreted as a model-discrimination experiment.

3. **Clinical efficacy remains empirical.** The conflict-coherence model predicts a *rank ordering* within a chosen endpoint, but it does not derive a unique biomarker, a dose-response curve, or a guaranteed clinical effect size. The magnitude of therapeutic benefit depends on biological factors (individual anatomy, pathology, attention) not captured by the mathematical model.

4. **The 40 Hz base rate is a single parameter.** The mode frequencies depend on the cycle rate $f_{\text{cycle}} = 40$ Hz. If the true base rate differs by more than $\sim$10%, the ladder would shift and predictions would need adjustment.

8.4 Future directions

Meditation protocols. The carrier theorem suggests designing meditation-entrainment protocols that stabilize theta first, then sequentially activate higher modes. Such protocols could be tested with real-time EEG-guided adaptive stimulation.

Compositional algorithms. The conflict-coherence functional provides an objective function for algorithmic music composition targeting hypothesized therapeutic endpoints: minimize conflict for relaxation, control conflict gradient for activation, target specific modes for band-selective entrainment.

Cross-modal extension. If the seven-mode basis extends to other sensory modalities (visual flicker, somatosensory vibration), coordinated multi-modal stimuli could produce stronger entrainment than any single channel.

9 Conclusion

We have presented a conditional framework in which neural oscillation structure is modeled as a seven-mode DFT-8 ladder, musical stimuli reach this ladder through low-frequency amplitude envelopes, a cost function ranks candidate stimuli within a phenomenological strain-proxy model, and the theta band occupies a mathematically unique position as the only full-ladder harmonic carrier. The framework combines one falsifiable structural hypothesis, one 40 Hz cycle calibration, and one candidate cost function (J-cost), and it produces eight specific experimental predictions together with study-design templates for refutation.

The goal of this paper is not to claim that music therapy "works" in any final or already-established sense—that remains an empirical question with a large and growing evidence base—but to provide a quantitative, falsifiable conditional model of *how* music could couple to a discrete oscillation ladder, which *specific* stimuli should be prioritized for testing within that model, and what structural role the theta band may play in the process. The internal mathematics is formally verified; the empirical interpretation remains open to confirmation or refutation. We invite direct experimental testing of the predictions in Table 5.

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